Microprice as a Martingale Corrector

A Unified Markov–Operator Framework for Fair Value and Value-Function Extensions

Lev Petersen

Abstract. Stoikov (2018) introduced the *microprice* as a horizon-free limit of conditional expectations of future prices, computed from Level-I order book data via an absorbing Markov chain. We reframe that construction as an operator statement about Markov additive processes at mid-price move times.

At move times the mid-price increment has a state-dependent conditional mean. We show that Stoikov's adjustment term is exactly the *Poisson corrector* (a Markov potential) on the post-move chain: the unique centered solution of a Poisson equation. Adding this corrector to the mid-price produces a martingale. The same Poisson equation yields martingale correctors for *any* additive order-book functional.

Many "fair value" objects can be expressed as value functions V solving $V = u + \Gamma M V$ for a Markov operator M and a continuation/discount operator Γ with $\|\Gamma M\| < 1$. In this regime the solution is a resolvent $V = (I - \Gamma M)^{-1} u$ with a convergent Neumann series. The microprice corresponds to the boundary case $\Gamma = I$: the resolvent becomes singular and well-posedness is recovered by centering. We provide clean existence/uniqueness conditions, operator-norm bounds for the absorbing-chain fundamental matrix, and truncation-error control for the main series solutions.

Keywords: microprice, martingale corrector, Poisson equation, Markov additive process, Markov potential, Neumann series, resolvent, absorbing Markov chain

Contents

1 Introduction **4**
1.1 Two guiding ideas . . . 4
1.2 Roadmap . . . 4

2 Setup and notation **5**
2.1 Prices and book features . . . 5
2.2 Event time vs. move time . . . 5
2.3 State discretization . . . 5
2.4 Standing assumptions . . . 5

3 Microprice mechanics: from absorbing event time to a Poisson equation **6**
3.1 Transition decomposition . . . 6
3.2 The fundamental matrix of the no-move kernel . . . 6
3.3 Post-move chain and first adjustment . . . 7
3.4 The microprice adjustment as a Poisson equation . . . 7
3.5 Microprice as a martingale corrector . . . 8

4 Microprice as a general martingale constructor **10**
4.1 Doob decomposition for state observables . . . 10
4.2 Poisson correctors for additive functionals . . . 10
4.3 Vector-valued functionals . . . 10

5 Value functions and Neumann-series resolvents **11**
5.1 Neumann-series value functions . . . 11
5.2 Microprice as a centered boundary case . . . 11

6 Discussion and limitations **12**

7 Conclusion **13**

A Proofs **14**

1 Introduction

Stoikov's microprice [1] starts from a practical microstructure question: given the current state of the limit order book, what is a fair estimate of the asset's value? The classical mid-price

$$p_t := \frac{b_t + a_t}{2}$$

ignores information in queue imbalance. A fair value should incorporate how the current book state predicts future mid-price moves.

1.1 Two guiding ideas

Our goal is to make explicit two ideas that are present but implicit in the original construction.

At *move times* (times when the mid-price changes), the mid-price forms a Markov additive process: the state evolves as a finite Markov chain and the price increment has a state-dependent conditional mean. Stoikov's "microprice adjustment" is the Poisson corrector that removes this predictable component, turning price-plus-corrector into a martingale. This viewpoint immediately generalizes: the same Poisson equation produces martingale correctors for any additive order-book functional.

Once the relevant Markov operator is identified, a broad class of "fair value" objects are value functions V solving a linear fixed-point equation of the form

$$V = u + \Gamma M V,$$

where M is a Markov operator and Γ encodes discounting/continuation. When $\|\Gamma M\| < 1$, the inverse $(I - \Gamma M)^{-1}$ exists and the solution is given by a convergent Neumann series. The microprice corresponds to the boundary case $\Gamma = I$, where the resolvent becomes singular and well-posedness is recovered by centering.

1.2 Roadmap

Section 2 fixes the event-time and move-time models and notation. Section 3 derives the post-move chain, the first adjustment, and the Poisson equation behind Stoikov's microprice. Section 4 elevates this to a general martingale-constructor theorem for additive order-book functionals. Section 5 introduces Neumann-series value functions and explains the microprice operator as a centered boundary case. Section 6 discusses modelling assumptions and practical conditioning issues. Proofs are collected in Appendix A.

2 Setup and notation

We work with Level-I order book data. At each event time t we observe the best bid b_t, best ask a_t, and their queue sizes B_t and A_t.

2.1 Prices and book features

We use the following derived quantities:

$$\begin{aligned}
\text{Mid-price:} \quad & p_t = (b_t + a_t)/2, \\
\text{Spread:} \quad & s_t = a_t - b_t, \\
\text{Imbalance:} \quad & i_t = \frac{B_t - A_t}{B_t + A_t} \in [-1, 1], \\
\text{Size-weighted mid (instantaneous “microprice” proxy):} \quad & m_t = \frac{a_t B_t + b_t A_t}{B_t + A_t} = p_t + i_t \cdot \frac{s_t}{2}.
\end{aligned} \tag{1}$$

The object of interest is not m_t itself, but rather a *fair value* obtained by conditioning on the book state and looking forward to future price moves.

2.2 Event time vs. move time

There are two natural clocks. Event time counts every book update (queue change, trade, etc.); the discretized state evolves as a Markov chain at this clock. Move time counts only events that change the mid-price, and this embedded chain is the right object for the microprice/martingale-corrector viewpoint.

Let $\tau_0 = 0$ and let

$$\tau_{k+1} := \inf\{t > \tau_k : p_t \neq p_{t-}\}$$

be the $(k+1)$-st mid-price move time. Define the move-time price and state

$$p_k := p_{\tau_k}, \qquad X_k := X_{\tau_k}.$$

2.3 State discretization

We discretize continuous features into buckets. For the single-instrument setting, a common choice is

$$X_t = (\text{spread bucket}, \text{imbalance bucket}),$$

with m spread buckets and n imbalance buckets, giving $L = mn$ states. Nothing in the operator algebra depends on this particular choice; any finite Markov state can be used.

2.4 Standing assumptions

Assumption 1 (Markov property)**.** The discretized state X_t is Markov at event times.

Assumption 2 (Bounded increments)**.** One-step mid-price changes satisfy $|\Delta p_t| \leq D$ for some constant D.

Assumption 3 (Ergodicity at move times)**.** The post-move chain (defined in Section 3) is irreducible and aperiodic, admitting a unique stationary distribution.

3 Microprice mechanics: from absorbing event time to a Poisson equation

This section makes explicit the mechanics behind Stoikov's microprice computation. We first decompose the event-time kernel into no-move and move pieces (Q, T), then use the fundamental matrix $(I - Q)^{-1}$ to compute the embedded post-move chain B and the one-move drift G_1, and finally recognize the infinite-horizon adjustment as the centered solution to a Poisson equation on B.

3.1 Transition decomposition

Partition event-time transitions according to whether the mid-price moves. Let Q_{xy} denote the probability of transitioning from state x to y without a price move, and T_{xy} the probability with a move. The full event-time transition matrix is

$$P := Q + T,$$

and is row-stochastic.

Let

$$p_{\text{move}}(x) := \sum_y T_{xy} = 1 - \sum_y Q_{xy}$$

be the probability the next event is a mid-price move.

Conditional on a move from state x, let R_{xj} denote the probability that the price change equals k_j (so $\sum_j R_{xj} = 1$), where $\{k_j\}_j$ is the finite set of possible move sizes. Let $\mathbf{k}$ be the column vector with entries k_j. The conditional expected change is

$$r_x^{\text{move}} := \sum_j R_{xj} k_j,$$

collected into the vector form $r^{\text{move}} = R\,\mathbf{k}$. The unconditional one-step drift (including zero change on no-move events) is

$$\bar{r}_x := p_{\text{move}}(x)\, r_x^{\text{move}}, \qquad \text{i.e.} \qquad \bar{r} = D_{p_{\text{move}}}\, r^{\text{move}},$$

where $D_{p_{\text{move}}} := \operatorname{diag}(p_{\text{move}}(x))$.

3.2 The fundamental matrix of the no-move kernel

The no-move matrix Q is substochastic: each row sums to at most 1. When moves occur with uniformly positive probability, the spectral radius of Q is strictly less than one, and the fundamental matrix exists.

Definition 3.1. A matrix $Q \in \mathbb{R}^{L \times L}$ is *substochastic* if $Q_{xy} \geq 0$ for all x, y and $\sum_y Q_{xy} \leq 1$ for all x.

Lemma 3.2 (Spectral radius bound)**.** *Let Q be substochastic with $\sum_y Q_{xy} \leq 1 - \varepsilon$ for all x. Then $\rho(Q) \leq 1 - \varepsilon < 1$.*

Proposition 3.3 (Fundamental matrix)**.** *Under the conditions of Lemma 3.2, the matrix $I - Q$ is invertible with*

$$N := (I - Q)^{-1} = \sum_{k=0}^{\infty} Q^k, \qquad \|N\|_\infty \le \frac{1}{\varepsilon}.$$

The entries N_{xy} equal the expected number of visits to y, starting from x, before the first price move.

Lemma 3.4 (Non-negativity)**.** *The fundamental matrix N has non-negative entries, with $N_{xx} \ge 1$ for all x.*

3.3 Post-move chain and first adjustment

Definition 3.5. The *first adjustment* is the expected cumulative price change up to the first move:

$$G_1 := (I - Q)^{-1}\bar{r} = N\,\bar{r} = N D_{p_{\text{move}}} r^{\text{move}}.$$

The *post-move chain* is the transition matrix observed only at move times:

$$B := (I - Q)^{-1}T = NT.$$

Lemma 3.6 (Embedded-chain interpretation)**.** *Let $(X_n)_{n\ge 0}$ be the event-time Markov chain with transition matrix $P = Q + T$, and let*

$$\tau := \inf\{n \ge 1 : \textit{the step } (X_{n-1} \to X_n) \textit{ is a price move}\}.$$

Then $B_{xy} = \mathbb{P}_x(X_\tau = y)$ is the distribution of the post-move state, and the first adjustment is

$$G_1(x) = \mathbb{E}_x[\Delta p_\tau],$$

the expected mid-price change at the next move. Equivalently, G_1 is the unique solution of the first-step system

$$G_1 = \bar{r} + QG_1. \tag{2}$$

Lemma 3.7. *If $Q + T$ is row-stochastic and $\rho(Q) < 1$, then B is row-stochastic.*

3.4 The microprice adjustment as a Poisson equation

Fix a move-time index k. For $n \ge 0$, define the *n-move fair value*

$$p_k^{(n)} := \mathbb{E}[p_{k+n} \mid X_k]\,.$$

Equivalently, define the *n-move adjustment* $G^{(n)} : \mathcal{X} \to \mathbb{R}$ by

$$G^{(n)}(x) := \mathbb{E}[p_{k+n} - p_k \mid X_k = x]\,,$$

so that $p_k^{(n)} = p_k + G^{(n)}(X_k)$. By time-homogeneity of the move-time dynamics, $G^{(n)}$ does not depend on k.

When the limit exists, the *microprice* (horizon-free fair value) is

$$p_k^{\text{micro}} := \lim_{n\to\infty} p_k^{(n)} = p_k + G^*(X_k), \qquad \text{where} \qquad G^* := \lim_{n\to\infty} G^{(n)}.$$

By the tower property and the Markov property at move times, the adjustments satisfy the recursion

$$G^{(0)} = 0, \qquad G^{(n)} = G_1 + BG^{(n-1)}.$$

Unrolling gives

$$G^{(n)} = \sum_{k=0}^{n-1} B^k G_1.$$

When it converges, the infinite-horizon adjustment is

$$G^* := \lim_{n\to\infty} G^{(n)} = \sum_{k=0}^{\infty} B^k G_1. \tag{3}$$

Since B is stochastic (hence has eigenvalue 1), the convergence of (3) requires a *centering* condition.

Definition 3.8 (Centering)**.** Let π be the stationary distribution of B. We say G_1 is *centered* if $\pi^T G_1 = 0$.

Theorem 3.9 (Microprice convergence)**.** *Let B be an ergodic Markov kernel on a finite state space with stationary distribution π. Let*

$$\lambda_* := \max\{|\lambda| : \lambda \in \sigma(B),\ \lambda \neq 1\} \in [0, 1)$$

be the second-largest eigenvalue modulus (SLEM). If G_1 is centered, i.e. $\pi^T G_1 = 0$, then:

1. *The series (3) converges absolutely to*

$$G^* = (I - B + \mathbf{1}\pi^T)^{-1} G_1.$$

   In particular, G^* is centered: $\pi^T G^* = 0$.

2. *G^* is the unique solution of the centered Poisson equation*

$$(I - B)G^* = G_1, \qquad \pi^T G^* = 0. \tag{4}$$

3. *For any $\rho \in (\lambda_*, 1)$ there exists $C_\rho < \infty$ such that the truncation error after n terms satisfies*

$$\left\| G^* - \sum_{k=0}^{n-1} B^k G_1 \right\|_\infty \leq \frac{C_\rho \rho^n}{1-\rho} \|G_1\|_\infty.$$

3.5 Microprice as a martingale corrector

At move times, (X_k, p_k) is a Markov additive process:

$$\mathbb{P}(X_{k+1} = y \mid X_k = x) = B_{xy}, \qquad \mathbb{E}[\Delta p_{k+1} \mid X_k = x] = G_1(x),$$

where $\Delta p_{k+1} := p_{k+1} - p_k$.

Theorem 3.10 (Microprice martingale)**.** *Assume the centered Poisson equation* (4) *has solution G^* (equivalently, $\pi^T G_1 = 0$). Then the move-time process*

$$\mathcal{P}_k := p_k^{\text{micro}} := p_k + G^*(X_k) \tag{5}$$

is a martingale with respect to the move-time filtration.

Remark 3.11 (Trending markets and discounted surrogates). If $\mu := \pi^T G_1 \neq 0$ (nonzero long-run drift per move), the undiscounted series (3) diverges linearly. One still has a martingale decomposition:

$$p_k - k\mu + \tilde{G}(X_k) \ \text{ is a martingale,}$$

where $\tilde{G}$ solves the centered Poisson equation $(I - B)\tilde{G} = G_1 - \mu\mathbf{1}$. In trending/regime-shifting markets, one should either remove the stationary drift (as in Theorem 4.1) or work with a well-posed value function satisfying a Neumann-series condition (Section 5).

Proposition 3.12 (First adjustment bounds)**.** *If the mid-price move size is uniformly bounded by $|\Delta p| \leq D$, then $\|G_1\|_\infty \leq D$.*

If one instead bounds via the operator norm, $\|G_1\|_\infty \leq \|N\|_\infty \|\bar{r}\|_\infty$, and imposes an absorption floor $p_{\text{move}}(x) \geq \varepsilon$ so that $\|N\|_\infty \leq 1/\varepsilon$, then $\|G_1\|_\infty \leq \|\bar{r}\|_\infty/\varepsilon$.

4 Microprice as a general martingale constructor

The microprice is one instance of a general principle: *additive* functionals driven by a Markov state admit a predictable drift term plus a martingale term, and the martingale term is obtained by solving a Poisson equation.

4.1 Doob decomposition for state observables

Let $(X_k)_{k\geq 0}$ be a Markov chain with transition matrix B. For any bounded function $f : \mathcal{X} \to \mathbb{R}$, the process $f(X_k)$ admits the decomposition

$$f(X_k) = f(X_0) + M_k + A_k, \tag{6}$$

where

$$A_{k+1} - A_k = (Bf - f)(X_k), \qquad M_{k+1} - M_k = f(X_{k+1}) - Bf(X_k),$$

and (M_k) is a martingale. Equation (6) is the discrete-time Doob–Meyer decomposition.

4.2 Poisson correctors for additive functionals

Now let $(Y_k)_{k\geq 0}$ be an *additive* functional driven by the chain:

$$Y_{k+1} = Y_k + \xi_{k+1}, \qquad \bar{\xi}(x) := \mathbb{E}[\xi_{k+1} \mid X_k = x].$$

The stationary mean drift is $\mu := \pi^T \bar{\xi}$. The centered drift is $h := \bar{\xi} - \mu\mathbf{1}$.

Theorem 4.1 (Markov additive martingale corrector)**.** *Assume B is ergodic with stationary distribution π, and let h be centered ($\pi^T h = 0$). Let g be the unique centered solution of the Poisson equation*

$$(I - B)g = h, \qquad \pi^T g = 0. \tag{7}$$

Then the compensated process

$$\mathcal{M}_k := Y_k - k\mu + g(X_k) - g(X_0) \tag{8}$$

is a martingale.

The microprice martingale (Theorem 3.10) is the special case $Y_k = p_k$ and $h = G_1 - \mu\mathbf{1}$.

Scope. Theorem 4.1 is purely structural: once an observable is written as an additive functional with conditional mean drift $\bar{\xi}(x)$, solving the Poisson equation (7) produces the drift-corrected martingale. We omit application-specific choices of ξ and focus on the operator construction.

4.3 Vector-valued functionals

All results extend componentwise to $\mathbb{R}^d$-valued rewards: one solves the same Poisson equation with multiple right-hand sides to obtain a vector martingale corrector.

5 Value functions and Neumann-series resolvents

The microprice correction G^* is an *undiscounted* potential on the post-move chain. It sits at the boundary of a more classical regime in which one introduces discounting/continuation so that the relevant inverse is given by a convergent Neumann series. This section records a general value-function construction and clarifies how the microprice operator appears as a centered boundary case.

5.1 Neumann-series value functions

Let $M \in \mathbb{R}^{L\times L}$ be a row-stochastic matrix (a Markov kernel on a finite state space $\mathcal{X}$). Let $\Gamma \in \mathbb{R}^{L\times L}$ be a bounded linear operator with

$$\|\Gamma M\|_\infty < 1, \tag{9}$$

where $\|\cdot\|_\infty$ denotes the induced ℓ^∞ operator norm. Given a bounded payoff/reward $u : \mathcal{X} \to \mathbb{R}$, consider the linear fixed-point equation

$$V = u + \Gamma M V. \tag{10}$$

This template covers a wide range of discounted or randomized-horizon objectives; the only property used below is the contraction condition (9).

Proposition 5.1 (Neumann-series value function)**.** *Assume* (9)*. Then* (10) *has a unique bounded solution given by*

$$V = (I - \Gamma M)^{-1} u = \sum_{k=0}^{\infty} (\Gamma M)^k u. \tag{11}$$

Moreover,

$$\|V\|_\infty \le \frac{1}{1 - \|\Gamma M\|_\infty}\, \|u\|_\infty.$$

Remark 5.2 (Same mechanism as the absorbing fundamental matrix)*.* The absorbing-chain fundamental matrix $N = (I - Q)^{-1}$ in Section 3 is a Neumann-series inverse of exactly the same type: absorption implies $\rho(Q) < 1$, hence $N = \sum_{k\ge 0} Q^k$. In contrast, the post-move chain B has eigenvalue 1, so the undiscounted inverse requires centering.

5.2 Microprice as a centered boundary case

For $\alpha \in (0, 1)$, the discounted potential $(I - \alpha B)^{-1}$ is well-defined on all of $\mathbb{R}^L$ and admits the Neumann series

$$(I - \alpha B)^{-1} v = \sum_{k=0}^{\infty} \alpha^k B^k v.$$

As $\alpha \uparrow 1$, the resolvent becomes singular because B has eigenvalue 1. On the centered subspace $\{v : \pi^T v = 0\}$, the limit exists and equals the microprice operator.

Proposition 5.3 (Resolvent limit)**.** *On the centered subspace* $\{v : \pi^T v = 0\}$*, the discounted resolvent* $(I - \alpha B)^{-1}$ *converges as* $\alpha \uparrow 1$ *to the microprice operator* $(I - B + \mathbf{1}\pi^T)^{-1}$*.*

6 Discussion and limitations

The literature often uses "microprice" both for the instantaneous size-weighted mid m_t and for the long-horizon fair value $p_k + G^*(X_k)$. Keeping these distinct avoids conceptual confusion.

The condition $\pi^T G_1 = 0$ asserts zero stationary drift per move and is substantive. If the drift is nonzero (trends, regime shifts), the undiscounted microprice is ill-posed; one should work with centered Poisson equations or with well-posed resolvents satisfying a Neumann-series condition (Section 5).

Bounds based on $\|N\|_\infty \leq 1/\varepsilon$ require a uniform lower bound on move probabilities. In practice, some states may exhibit extremely rare moves, making N ill-conditioned and amplifying estimation error.

The framework is operator-theoretic: it applies to whatever finite-state Markov approximation is used. If the chosen state is not approximately Markov, the resulting kernels are unstable and predictive performance degrades; this is an empirical modelling issue rather than a limitation of the algebra.

7 Conclusion

Stoikov's microprice can be read as an operator identity rather than a collection of model variants. At mid-price move times, the mid-price is a Markov additive process; the microprice adjustment is the Poisson corrector on the post-move chain. Adding this corrector to the price produces a martingale, and the same Poisson equation constructs martingales for general additive order-book functionals.

A second message is that many "fair value" constructions are value functions for the same Markov skeleton. When the relevant operator inverse satisfies a Neumann-series condition, the value function is a well-posed resolvent. The microprice corresponds to the boundary case where the eigenvalue at 1 forces centering; the microprice operator is the centered limit of discounted resolvents.

Empirical validation and model-selection choices for the Markov state are left to future work.

A Proofs

Proof of Lemma 3.2

For any matrix A, the spectral radius satisfies $\rho(A) \leq \|A\|$ in any submultiplicative operator norm. Taking the induced ℓ^∞ norm,

$$\|Q\|_\infty = \max_x \sum_y |Q_{xy}| = \max_x \sum_y Q_{xy} \leq 1 - \varepsilon.$$

Hence $\rho(Q) \leq 1 - \varepsilon < 1$. □

Proof of Proposition 3.3

Since $\rho(Q) < 1$, the Neumann series $\sum_{k=0}^{\infty} Q^k$ converges absolutely and equals $(I-Q)^{-1}$. For the norm bound,

$$\|N\|_\infty \leq \sum_{k=0}^{\infty} \|Q^k\|_\infty \leq \sum_{k=0}^{\infty} \|Q\|_\infty^k \leq \sum_{k=0}^{\infty} (1-\varepsilon)^k = \frac{1}{\varepsilon}.$$

The probabilistic interpretation is standard for absorbing chains: N_{xy} is the expected number of visits to y before absorption when starting from x. □

Proof of Lemma 3.4

From $N = I + Q + Q^2 + \cdots$ and non-negativity of Q, each term is non-negative entrywise. Thus $N \geq 0$ and $N_{xx} \geq 1$ for all x. □

Proof of Lemma 3.6

Let τ be the first event-time index at which a mid-price move occurs, as defined in Lemma 3.6. Conditioning on the first step from x gives the first-step equation

$$G_1(x) = \sum_y Q_{xy} G_1(y) + \sum_y T_{xy}\, \mathbb{E}[\Delta p \mid x \to y,\ \text{move}].$$

By definition, the second term is $\bar{r}_x$, hence $G_1 = \bar{r} + QG_1$ and $(I-Q)G_1 = \bar{r}$. Since $\rho(Q) < 1$, $I - Q$ is invertible and $G_1 = (I-Q)^{-1}\bar{r}$.

For the embedded chain, note that

$$\mathbb{P}_x(X_\tau = y) = \sum_{k=0}^{\infty} \mathbb{P}_x(X_k = y,\ \tau = k+1) = \sum_{k=0}^{\infty} (Q^k T)_{xy} = \big((I-Q)^{-1}T\big)_{xy} = B_{xy}.$$

□

Proof of Lemma 3.7

From $(Q+T)\mathbf{1} = \mathbf{1}$, we have $T\mathbf{1} = (I-Q)\mathbf{1}$. Thus

$$B\mathbf{1} = (I-Q)^{-1}T\mathbf{1} = (I-Q)^{-1}(I-Q)\mathbf{1} = \mathbf{1}.$$

Non-negativity of B follows from non-negativity of N and T. □

Proof of Theorem 3.9

Write

$$B = \mathbf{1}\pi^T + C, \qquad C := B - \mathbf{1}\pi^T.$$

Since $B\mathbf{1} = \mathbf{1}$ and $\pi^T B = \pi^T$, we have $C\mathbf{1} = 0$ and $\pi^T C = 0$. Ergodicity implies that 1 is a simple eigenvalue of B and all other eigenvalues have modulus < 1, so $\rho(C) = \lambda_* < 1$.

Centering gives $\mathbf{1}\pi^T G_1 = 0$, hence for all $k \geq 0$,

$$B^k G_1 = (\mathbf{1}\pi^T + C)^k G_1 = C^k G_1.$$

Therefore,

$$\sum_{k=0}^{\infty} B^k G_1 = \sum_{k=0}^{\infty} C^k G_1 = (I - C)^{-1} G_1,$$

which proves the rank-one inverse formula since $I - C = I - B + \mathbf{1}\pi^T$. Moreover, $\pi^T B^k = \pi^T$ for all k, so $\pi^T G^* = \sum_{k\geq 0} \pi^T G_1 = 0$ and G^* is centered.

To verify the Poisson equation, start from the partial sums $S_n := \sum_{k=0}^{n-1} B^k G_1$. Then

$$(I - B)S_n = G_1 - B^n G_1.$$

Since $B^n G_1 = C^n G_1 \to 0$ as $n \to \infty$ (because $\rho(C) < 1$), taking $n \to \infty$ yields $(I - B)G^* = G_1$. Uniqueness under the centering condition follows because the nullspace of $I - B$ is $\mathrm{span}\{\mathbf{1}\}$: if $(I - B)g = 0$ and $\pi^T g = 0$, then g must be constant and hence $g \equiv 0$.

For truncation, note that $G^* - S_n = \sum_{k=n}^{\infty} C^k G_1$. In finite dimensions, for any $\rho \in (\rho(C), 1)$ there exists $C_\rho < \infty$ such that $\|C^k\|_\infty \leq C_\rho \rho^k$ for all k. Hence

$$\|G^* - S_n\|_\infty \leq \sum_{k=n}^{\infty} \|C^k\|_\infty \, \|G_1\|_\infty \leq \sum_{k=n}^{\infty} C_\rho \rho^k \|G_1\|_\infty = \frac{C_\rho \rho^n}{1 - \rho} \|G_1\|_\infty.$$

□

Proof of Theorem 3.10

Using the Markov property at move times,

$$\begin{aligned}
\mathbb{E}[\mathcal{P}_{k+1} \mid \mathcal{F}_k] &= \mathbb{E}[p_{k+1} + G^*(X_{k+1}) \mid \mathcal{F}_k] \\
&= p_k + \mathbb{E}[\Delta p_{k+1} \mid X_k] + \mathbb{E}[G^*(X_{k+1}) \mid X_k] \\
&= p_k + G_1(X_k) + (BG^*)(X_k).
\end{aligned}$$

From $(I - B)G^* = G_1$ we have $G_1 + BG^* = G^*$ pointwise, hence $\mathbb{E}[\mathcal{P}_{k+1} \mid \mathcal{F}_k] = p_k + G^*(X_k) = \mathcal{P}_k$. □

Proof of Theorem 4.1

Let $\mathcal{M}_k$ be defined by (8). Then

$$\mathcal{M}_{k+1} - \mathcal{M}_k = \xi_{k+1} - \mu + g(X_{k+1}) - g(X_k).$$

Taking conditional expectation given $\mathcal{F}_k$ and using $\mathbb{E}[\xi_{k+1} \mid X_k] = \bar{\xi}(X_k)$ and $\mathbb{E}[g(X_{k+1}) \mid X_k] = Bg(X_k)$ gives

$$\mathbb{E}[\mathcal{M}_{k+1} - \mathcal{M}_k \mid \mathcal{F}_k] = (\bar{\xi} - \mu\mathbf{1})(X_k) + (Bg - g)(X_k) = h(X_k) - (g - Bg)(X_k) = 0,$$

where the last equality is (7) written as $g - Bg = h$. Hence $(\mathcal{M}_k)$ is a martingale. □

Proof of Proposition 5.1

Assume $\|\Gamma M\|_\infty < 1$. Then the Neumann series $\sum_{k=0}^{\infty}(\Gamma M)^k$ converges absolutely in the induced operator norm, and defines a bounded linear operator

$$S := \sum_{k=0}^{\infty}(\Gamma M)^k.$$

Telescoping gives

$$(I - \Gamma M)S = \sum_{k=0}^{\infty}(\Gamma M)^k - \sum_{k=1}^{\infty}(\Gamma M)^k = I,$$

so $S = (I - \Gamma M)^{-1}$. Setting $V = Su$ yields (11), and multiplying by $(I - \Gamma M)$ gives the fixed point equation (10).

For uniqueness, if V and W both solve (10), then $V - W = \Gamma M(V - W)$. Taking $\|\cdot\|_\infty$ norms gives $\|V - W\|_\infty \le \|\Gamma M\|_\infty \|V - W\|_\infty$, hence $V = W$.

Finally,

$$\|V\|_\infty \le \sum_{k=0}^{\infty} \|\Gamma M\|_\infty^k \, \|u\|_\infty = \frac{1}{1 - \|\Gamma M\|_\infty} \, \|u\|_\infty.$$

□

Proof of Proposition 5.3

Let v satisfy $\pi^T v = 0$ and write $B = \mathbf{1}\pi^T + C$ with $C = B - \mathbf{1}\pi^T$ as above. Then $B^k v = C^k v$ for all $k \ge 0$. For $\alpha \in (0, 1)$,

$$(I - \alpha B)^{-1} v = \sum_{k=0}^{\infty} \alpha^k B^k v = \sum_{k=0}^{\infty} \alpha^k C^k v = (I - \alpha C)^{-1} v.$$

Since $\rho(C) < 1$, we have $(I - \alpha C)^{-1} \to (I - C)^{-1}$ as $\alpha \uparrow 1$. Finally, $I - C = I - B + \mathbf{1}\pi^T$, so the limit is $(I - B + \mathbf{1}\pi^T)^{-1} v$. □

References

[1] S. Stoikov, "The Micro-Price: A High Frequency Estimator of Future Prices," *Quantitative Finance*, vol. 18, no. 12, pp. 1959–1966, 2018.